WORKED SOLUTION

STEP II 1990, Question 13

Mechanics • Equilibrium

STEP II 1990, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q13

The key idea

Because the peg is smooth and the string light, the tension is the same all the way along the string, so the rod feels two *equal* pulls, along PO and along QO. Their resultant bisects the angle POQ, and for equilibrium it must balance the weight along the same vertical line. So the vertical through O is the angle bisector of POQ and passes through G, and the angle bisector theorem does the rest. The inequality is then just the statement that the triangle OPQ is a genuine triangle.

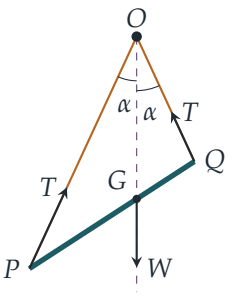
Solution

Finding OP and OQ

Model the rod as a rigid rod with centre of gravity G, where $PG = a + d$ and $GQ = a - d$; we take $0 \leq d < a$, so that G lies on the rod, on the Q side of the midpoint (or at it). The string is light and inextensible and the peg is small and smooth, so the tension T is the same in both parts OP and OQ, and

$$OP + OQ = 2b.$$

The rod is acted on by just three forces: its weight W at G, a tension T at P along PO, and a tension T at Q along QO.



The lines of action of the two tensions meet at O, so (taking moments about O) the weight must also act through O: the vertical through O passes through G. Moreover, the two tensions have equal magnitude, so their resultant points along the sum of two unit vectors, one along PO and one along QO, which is the direction of the bisector of the angle between them. This resultant must be vertical to balance W. Hence the vertical line OG bisects the angle POQ; call each half α.

Ask yourself What does knowing that OG bisects the angle POQ tell us about the lengths?

By the angle bisector theorem in triangle OPQ, the bisector from O divides PQ in the ratio of the adjacent sides:

$$\frac{OP}{OQ} = \frac{PG}{GQ} = \frac{a+d}{a-d}.$$

(To see this directly: triangles OPG and OQG share the height from O, so their areas are in the ratio $PG : GQ$; they also have areas $\frac{1}{2}\,OP \cdot OG \sin\alpha$ and $\frac{1}{2}\,OQ \cdot OG \sin\alpha$.) So $OP = k(a+d)$ and $OQ = k(a-d)$ for some $k > 0$, and $OP + OQ = 2b$ gives $2ka = 2b$, so $k = b/a$:

$$OP = \frac{b(a+d)}{a}, \qquad OQ = \frac{b(a-d)}{a}.$$

The inequality

Let $\theta = \angle POQ = 2\alpha$. Because the rod is not vertical, O is not on the line PQ, so OPQ is a genuine triangle and $0 < \theta < \pi$. The cosine rule with $PQ = 2a$ gives

$$\begin{aligned} 4a^2 &= OP^2 + OQ^2 - 2\,OP \cdot OQ \cos\theta \\ &= \frac{b^2}{a^2}\Big[(a+d)^2 + (a-d)^2 - 2(a^2-d^2)\cos\theta\Big] \\ &= \frac{2b^2}{a^2}\Big[(a^2+d^2) - (a^2-d^2)\cos\theta\Big]. \end{aligned}$$

Now $\cos\theta < 1$ because $\theta \neq 0$, and $a^2 - d^2 > 0$ because $d < a$. So

$$4a^2 > \frac{2b^2}{a^2}\Big[(a^2+d^2) - (a^2-d^2)\Big] = \frac{4b^2d^2}{a^2},$$

that is, $b^2d^2 < a^4$. Taking positive square roots (with $b > 0$, $d \geq 0$), $bd < a^2$, so

$$d < \frac{a^2}{b},$$

as required.

Equivalently, this is the triangle inequality $OP - OQ < PQ$: here $OP - OQ = 2bd/a$, and $2bd/a < 2a$ is the same condition. (The other inequality, $\cos\theta > -1$, gives $OP + OQ > PQ$, i.e. $b > a$, which just says that the string is long enough to go over the peg at all.)

Checks

- If $d = 0$ (uniform rod), $OP = OQ = b$ and the rod hangs horizontally, as symmetry demands; the inequality $0 < a^2/b$ holds automatically.
- As $d \to a^2/b$, $OP - OQ \to 2a = PQ$, so the triangle flattens and the rod becomes vertical: beyond this value the only way to hang is vertically.
- With $a = 1$, $b = 2$, $d = 0.3$: $OP = 2.6$, $OQ = 1.4$, $\cos\theta \approx 0.648$. Placing this triangle so that OG is vertical, the horizontal components of the two tensions cancel and the moments about O balance to machine precision. A scan over all positions of the string confirmed this is the only non-vertical equilibrium, and that none exists once $d \geq a^2/b$.

Common slips

- Assuming the tensions in the two parts of the string differ. Over a smooth peg they are equal, and that is what makes OG the angle bisector.
- Using the angle bisector ratio the wrong way round: the longer string, OP, goes to the end further from G.
- Expecting this equilibrium to be stable. In fact it is unstable: with $d = 0$ the horizontal rod has G a distance $\sqrt{b^2 - a^2}$ below O, while hanging vertically puts it a distance b below, which is lower. The question asks only about equilibrium.

Beyond STEP

The locus of O relative to the rod is an ellipse with foci P and Q, and the equilibrium positions are where the height of G is stationary. That is the principle of virtual work, which leads on to Lagrangian mechanics. It also explains why equal tensions bisect the angle: it is the reflection property of the ellipse. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good next step for statics and stability.

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